

MODEL LAB

CSA05 – Database Management Systems - Practical Examinations Questions Paper

Time: 3 Hours

SET -6

Max Marks: 100 (25 each)

Banking & Financial System

Scenario

A commercial bank tracks user accounts and financial transactions.

- Customers (cust_id, name, phone, city)
- Accounts (account_no, cust_id, account_type, balance)
- Transactions (txn_id, account_no, txn_type, amount, txn_date)
- AuditLog (log_id, account_no, old_balance, new_balance, changed_at)

Questions

1. **SQL & Joins:** Write a query using SELF JOIN or subqueries to list accounts that have a higher balance than the average balance of all accounts in the same account_type.
2. **Views:** Create a view vw_account_summary that lists customer name, account number, account type, and total count of transactions performed per account.
3. **User-Defined Function:** Create a function fn_calculate_interest that takes account_no and years as input, calculates simple interest at a rate of 5% for savings accounts, and returns the computed interest amount.
4. **Triggers:** Write an AFTER UPDATE trigger on the Accounts table that records the account_no, old_balance, new_balance, and execution timestamp into the AuditLog table whenever a balance changes.

```
mysql> SELECT *
-> FROM Accounts a
-> WHERE balance > (
->     SELECT AVG(balance)
->     FROM Accounts b
->     WHERE b.account_type = a.account_type
-> );
+-----+-----+-----+-----+
| account_no | cust_id | account_type | balance |
+-----+-----+-----+-----+
|      1001 |      101 | Savings      | 50000.00 |
+-----+-----+-----+-----+
1 row in set (0.05 sec)
```

```
mysql> CREATE VIEW vw_account_summary AS
-> SELECT c.name,
->         a.account_no,
->         a.account_type,
->         COUNT(t.txn_id) AS total_transactions
-> FROM Customer c
-> JOIN Accounts a
-> ON c.cust_id = a.cust_id
-> LEFT JOIN Transactions t
-> ON a.account_no = t.account_no
-> GROUP BY c.name, a.account_no, a.account_type;
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> SELECT * FROM vw_account_summary;
+-----+-----+-----+-----+
| name | account_no | account_type | total_transactions |
+-----+-----+-----+-----+
| Indhu | 1001 | Savings | 2 |
| Ramesh | 1002 | Savings | 1 |
| Priya | 1003 | Current | 1 |
+-----+-----+-----+-----+
3 rows in set (0.06 sec)
```

```
mysql> SELECT c.name,
->         t.account_no,
->         t.txn_type,
->         t.amount
-> FROM Customer c
-> JOIN Accounts a
-> ON c.cust_id = a.cust_id
-> JOIN Transactions t
-> ON a.account_no = t.account_no
-> WHERE t.amount > 5000;
+-----+-----+-----+-----+
| name | account_no | txn_type | amount |
+-----+-----+-----+-----+
| Indhu | 1001 | Deposit | 10000.00 |
| Ramesh | 1002 | Deposit | 8000.00 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT c.name,
->         a.account_no,
->         t.txn_type,
->         t.amount
-> FROM Customer c
-> JOIN Accounts a
-> ON c.cust_id = a.cust_id
-> JOIN Transactions t
-> ON a.account_no = t.account_no;
+-----+-----+-----+-----+
| name | account_no | txn_type | amount |
+-----+-----+-----+-----+
| Indhu | 1001 | Deposit | 10000.00 |
| Indhu | 1001 | Withdraw | 5000.00 |
| Ramesh | 1002 | Deposit | 8000.00 |
| Priya | 1003 | Withdraw | 2000.00 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```